How Much Does Time Matter?
A Comparative Study in Real-World Temporal Networks

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Abstract

Real networks are made of interactions that happen at specific times, yet they are routinely analysed as static aggregated graphs. We ask how much this discards. We organise the temporal-network literature around four questions—representation, activity patterns, structural evolution and function—distil seven testable hypotheses (H1–H7), and define a common pipeline of window analysis, randomized reference models and dynamic-community tracking on three networks of different type and time scale: face-to-face contacts in a primary school, online messaging among students, and a computer-science citation network. The cross-network comparison uses normalised time effects *rather than raw metrics. The completed citation-network study contains 19,150 papers and 73,269 strictly older-target citations. It supports sublinear degree preference and age-dependent citation choice, with a densification exponent of 1.343, but no detected shrinkage of the sampled effective diameter. A three-parameter citation model fits selected histories well without an established MAE advantage over linear extrapolation. In contacts and messages, window choice changes structural metrics and time ordering reduces reachability. Event-started SIR slows relative to timestamp shuffle in primary PS settings but peaks earlier in CM. These hypothesis-specific effects do not establish one overall ranking of network types.*

1. Introduction

Real networks evolve: nodes and links appear and disappear, and interactions take place at specific moments—a conversation lasts a minute, a message is sent at a given second, a paper cites earlier work in a given year. Classical network analysis nevertheless collapses all of this into one static graph in which two nodes are either linked or not. Such aggregation is convenient, but it discards when links are active, how often they recur, in which order they occur and how the network changes. Unlike for static graphs, connectedness is not transitive in time, so a temporal network cannot be reduced to a simple graph without loss [11]. The temporal-network literature [11, 12, 21] has shown that each of these ingredients can change the conclusions drawn about structure, communities, and processes such as epidemics.

Most of this evidence, however, comes from a single dataset or a single method. It is therefore hard to tell *how much* time matters in general, and for which kinds of systems. We address two research questions. **RQ1:** how does temporal information change our understanding of the structure and function of real networks, and how does this effect differ across network types? **RQ2:** which hypotheses from the temporal-network literature (H1–H7, Tab. 2) hold in which systems?

Approach. The report is organised around four questions that recur in every section (Fig. 1): (I) *representation*—what is lost by aggregation and by the choice of time window? (II) *activity patterns*—are interactions bursty, repeated and persistent? (III) *structural evolution*—how do networks grow, age and reorganise? (IV) *function*—does time ordering change reachability, centrality and spreading? Each question is turned into a literature hypothesis, a metric and a randomized reference model [6], and applied to three networks that differ in edge meaning, generating mechanism and time scale (seconds, seconds, years): face-to-face contacts, online messages and citations. Because raw metrics are not comparable across such systems, datasets are compared via *normalised time effects* (Sec. 4.7). We do not propose new models; the contribution lies in the common framework and the comparison.

Contributions. (1) A question-driven synthesis of how network scientists study temporal networks; (2) a common, reproducible pipeline that applies it to three types of real data; (3) a cross-network verdict on H1–H7, stating where time information is essential and where a static view suffices.



Figure 1. Structure of the study. Four questions (Parts I–IV) map to seven hypotheses and four method groups; the same order is used in Related Work, Methods, Results and Discussion. The three dataset types share a common workflow; P1/P2 apply to PS/CM, while citation-target controls P3 apply to CIT.

Roadmap. Section 2 reviews related work, Sec. 3 describes the data, Sec. 4 the methods, Sec. 5 the results, Sec. 6 discusses them and Sec. 7 concludes.

2. Related Work

We organise the literature by question rather than paper by paper; the subsections follow Parts I–IV of the Introduction. General frameworks and reviews are given by Holme [11], Holme and Saramäki [12] and the book of Masuda and Lambiotte [21]. A central point of these reviews is that the basic unit is a *contact* (two nodes and a time), and that connectedness is no longer transitive: if a meets b and later b meets c, a path from a to c exists, but not the other way round. Hence a temporal network cannot be turned into a simple graph without losing information or changing what a node means, and many static-network methods need nontrivial generalisations [11].

2.1. Representing time: aggregation, snapshots and windows

Temporal networks can be stored as contact sequences, interval graphs or sequences of snapshots [12]. Lossy representations such as weighted aggregates or reachability graphs reduce the data to a static object; lossless ones keep all contacts [11]. Each choice involves a trade-off: snapshots make static tools applicable but require a window length, whereas the full event sequence keeps all information. The effect of this choice has been quantified for communication data: window size and placement change the structure of the aggregated network [17], and coarser temporal resolution distorts properties of the time-varying network in a way that depends on the system [30]. This motivates **H1**: aggregation and window choice change structural conclusions.

2.2. Temporal activity patterns

Human activity is bursty: inter-event times are heavy-tailed rather than exponential [1], and consecutive intervals are correlated, which Goh and Barabási [8] capture with a burstiness coefficient and a memory coefficient. Social ties are also repeated and persistent over long periods [33], while the set of active links turns over. Temporal motifs [16, 27] and activity-driven models [28] describe and generate such patterns. This gives **H2** (activity is bursty and has memory) and **H3** (ties are repeated and persistent while the network turns over).

2.3. Structural evolution: growth, ageing and communities

Growth is classically explained by preferential attachment [2], whose strength can be measured directly from growth data [13, 24]. Real systems usually show ageing, where old nodes attract fewer new links [5], and in citations this is captured by a relevance decay [37]. Growing networks also densify, with the number of edges growing superlinearly in the number of nodes [18, 19]; node and link deletion alter these laws [22]. This gives **H4**: growth follows preferential attachment with ageing, and networks densify.

At the mesoscale, multilayer modularity couples partitions across time [23], communities can be tracked across snapshots through matching [9] and change can be mapped between partitions [32]; Rossetti and Cazabet [31] survey the field. Per-snapshot partitions are typically obtained with Louvain [3] or Leiden [36]. This gives **H5**: communities reorganise over time and a static partition hides this.

2.4. Time and network function

Because interactions occur in sequence, only time-respecting paths can carry information: reachability in contact sequences is much lower than in the aggregated graph [10], and path lengths and centrality change accordingly [15, 25]. Burstiness and temporal correlations further slow down spreading even when the static topology is unchanged [14]. These results give **H6** (time ordering restricts reachability and changes centrality rankings) and **H7** (burstiness and temporal correlations slow down spreading).

2.5. Randomized reference models

Randomization is more versatile for temporal than for static networks, where it is essentially limited to degree-

Type	Dataset	N	Events	Span	Res.
Contact	Primary school (PS)	242	125 773	2 d	20 s
Comm.	CollegeMsg (CM)	1 899	59 798	193 d	1 s
Citation	DBLP–DB (CIT)	19 150	73 269	50 yr	1 yr

Table 1. Data cards. N: nodes; Res.: temporal resolution. CIT spans publication years 1970–2020, with primary analysis in 1990–2020. PS: undirected, class labels, CC BY-NC-SA; CM: directed, no labels; CIT: directed, acyclic.

preserving link shuffling, because one can switch off individual kinds of correlation [11]. Randomized reference models act as hypothesis tests: each one preserves some properties of the data and destroys others, so differences between data and model reveal which feature matters [6]. We use them in Parts II–IV (Sec. 4.6).

Time scales and network types. Temporal networks are usually defined as systems whose evolution is not much slower than the processes running on them, and a single system contains many time scales (seconds, hours, days) [11]. Contact and communication data are event sequences with such a hierarchy, whereas citation networks are acyclic by construction and all out-links of a paper appear at one instant [11]. This is why we expect, and test, different sensitivity to time across network types.

Gap. Most studies examine one dataset or one method. Few compare, within a single framework, how sensitive different *types* of networks are to temporal information. The hypotheses distilled above (Tab. 2) allow such a comparison.

3. Data

We use one network per type, chosen to differ in edge meaning, generating mechanism and time scale (seconds, seconds, years); data cards are in Tab. 1 and activity time series in Fig. 2.

Face-to-face contact. The primary-school data [7, 34] were collected by the SocioPatterns collaboration with wearable proximity sensors over two school days. A contact is recorded when two people are face to face within about 1.5 m for at least 20 s. Each individual carries a class label, which we use as ground truth for communities. The data are released under CC BY-NC-SA.

Online communication. CollegeMsg [26] contains time-stamped private messages between 1 899 users of an online student community over 193 days at 1 s resolution. Edges are directed, and there are no node labels.

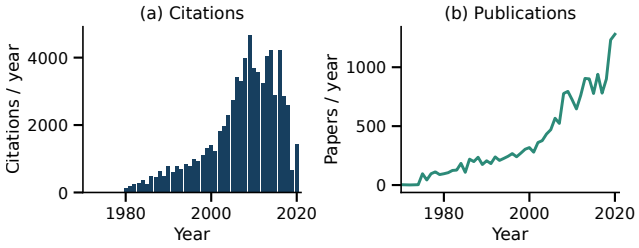


Figure 2. Measured CIT activity: (a) retained citations and (b) selected publications. The late citation decline despite rising publications motivates coverage checks. PS/CM activity series remain pending.

Scientific publications. We use AMiner DBLP-Citation V14 [35], selected by a fixed set of database venue families and their declared name variants. Streaming 5,259,858 records yields 19,150 publications from 1970–2020 and 73,269 distinct citations to strictly older selected papers. These are within-subgraph citations, not global citation counts; 6,709 nodes are isolates. We save 5,486 same-year references separately, exclude 6,390 later-year references and 494 self-loops, and remove 37,440 repeated reference entries. The graph is acyclic under the strict year rule. AMiner specifies research use; no additional open-data licence is inferred. The recent fall in observed references despite growing paper counts motivates a cutoff-2015 sensitivity check and cautions about coverage and censoring.

Preprocessing. All data are converted to a common event format (u, v, t) with integer time stamps (seconds or years) plus a table of node labels. Self-loops and exact duplicate events are removed. PS/CM windows retain the final partial window and include empty windows. Each source has known limits: citation data have year-level resolution, the contact data cover only two days, and the sensors sample only participants who wear them. The three datasets differ in edge meaning (proximity, message, citation), directedness and time scale by design, so that agreement across them is not an artefact of one data type.

4. Methods

Every analysis follows the same template: hypothesis from the literature → metric → randomized reference model. Table 2 lists the seven hypotheses, the methods (M1–M4) that test them and the datasets where each method is meaningful (19 method–dataset pairs). The empty cells are deliberate: CIT is a year-resolution DAG without fine-grained events (no M2a, M2b, M3) and without community reorganisation (no M4), while PS and CM have no growth process (no M2c).

Part	ID	Hypothesis	Source	Method	Null	PS	CM	CIT
I	H1	Aggregation and window choice change structural conclusions	[17, 30]	M1	–	✓	✓	✓
II	H2	Activity is bursty and has memory	[8]	M2a	P1	✓	✓	
	H3	Ties are repeated and persistent; the network turns over	[33]	M2b	P1	✓	✓	
III	H4	Growth follows preferential attachment with ageing; networks densify	[13, 18, 37]	M2c-1–4	P3			✓
	H5	Communities reorganise over time; the static partition hides this	[9, 23]	M4	P1	✓	✓	
IV	H6	Time ordering restricts reachability and changes centrality rankings	[10, 15, 25]	M3a, M3b	P1, P2	✓	✓	
	H7	Burstiness and temporal correlations slow down spreading	[14]	M3c	P1, P2	✓	✓	

Table 2. Hypotheses tested in this work, with methods, null models (Sec. 4.6) and applicable datasets.

4.1. Notation

Let $\mathcal{E} = \{(u, v, t)\}$ be the event sequence. The static aggregate G contains a link (u, v) if at least one event occurs between u and v. A snapshot $G_{\Delta t}(k)$ contains the events in $[k\Delta t, (k+1)\Delta t)$, and the cumulative network $G(\leq t)$ contains all events up to t.

4.2. Aggregation and time windows (M1)

For each window length Δt on a grid (PS: 5 min to 1 day; CM: 1 h to 1 month) we compute density, mean degree, clustering coefficient and giant-component size of the snapshots and of the aggregate. For CIT, yearly and cumulative networks are compared. *Window sensitivity* is $(\max m - \min m)/\overline{m}$ across window-width means. PS/CM use a fixed node universe for each selected period, include empty windows and retain the last partial window; windows start at the first observed event. Rankings are handled separately in M3b. Aggregation can introduce paths that cannot be traversed in time: if c–d occurs before a–b and b–c, the static path a–b–c–d is not time-respecting from a.

4.3. Activity patterns (M2a, M2b)

M2a: Burstiness and memory. We analyse inter-event times for each interacting node pair in the Primary School (PS) and CollegeMsg (CM) networks. PS contacts are recorded every 20 seconds. Consecutive records for the same unordered pair, separated by exactly 20 seconds, are merged into a single contact episode. For CM, each retained directed message is treated as an independent event.

For each edge, inter-event intervals are calculated from consecutive event timestamps. We pool these intervals across edges and calculate burstiness as

$$B = \frac{\sigma - \mu}{\sigma + \mu}, \tag{1}$$

where μ and σ are the mean and population standard deviation of the pooled intervals.

Memory is defined as the Pearson correlation between consecutive inter-event intervals within each edge, pooled across all valid adjacent interval pairs [8]. Both B and M are pooled statistics rather than averages of per-edge coefficients.

M2b: Repetition, lifetime and turnover. We quantify temporal link persistence using three metrics.

The repetition rate is the fraction of distinct edges with at least two independent contact episodes (PS) or two messages (CM).

Edge lifetime is defined as the difference between the last and first event timestamps of each edge. Edges observed only once have zero lifetime.

Link turnover is measured using the Jaccard similarity between edge sets in consecutive time windows:

$$J(E_k, E_{k+1}) = \frac{|E_k \cap E_{k+1}|}{|E_k \cup E_{k+1}|}. \tag{2}$$

We average this quantity over adjacent window pairs with nonempty unions.

The default window size is one hour for PS and seven days for CM. We additionally evaluate 15, 30, 60 and 120-minute windows for PS, and 1, 3, 7 and 14-day windows for CM.

For PS, edge lifetime and window Jaccard are computed from the original event stream. We also compare raw 20-second records with merged contact episodes to assess the sensitivity of H2 to event definition.

4.4. Structural evolution (M2c, M4)

Citation data and time resolution. We stream AMiner DBLP-Citation V14 [35] and retain an induced subgraph of database venues selected by a fixed whitelist (SIGMOD, VLDB/PVLDB, ICDE, PODS, EDBT, ICDT, TODS, VLDB Journal, Data & Knowledge Engineering, and Information Systems). All selected papers published in 1960–2020 are retained, including isolates. A distinct citation $u \to v$ occurs in the publication year of u. The

primary analysis requires $y_v < y_u$; same-year citations are saved for structural sensitivity analysis because their order cannot be resolved. We exclude self-citations to the same paper, duplicate edges and future references. Author self-citations are not removed. Counts refer to citations within the selected venues, rather than global citation impact. The primary analysis years are 1990–2020, initialized using the available earlier history.

M1: annual and cumulative citation structure. Annual and cumulative networks use the same node denominator: all selected papers already published by year t. Directed density is $E/[N(N-1)]$ and mean total degree is $2E/N$. Clustering, component counts and largest-component fractions use the weak undirected projection, including isolates. Trailing windows of 1, 2, 5 and 10 years provide a window-sensitivity check; sensitivity is $(\max - \min)/\text{mean}$ over the grid at each endpoint, then averaged over endpoints.

M2c-1: preferential attachment. Following the degree-class approach of Jeong et al. [13], we freeze in-degree at the start of each citing year and count eligible nodes n_{tk} and new citations h_{tk} in each prior-degree class k. We estimate α by conditional likelihood with probabilities proportional to $n_{tk}(k+1)^{\alpha}$, conditioning on the total number of citations in each year. The offset admits citations to previously uncited papers; offsets 0.5 and 2 provide sensitivity checks. Confidence intervals use 400 resamples of citing-year blocks. These are descriptive uncertainty intervals; neither positive α nor its departure from an age-only null identifies a causal mechanism.

M2c-2: ageing. For ages 1–60, the target probability in citing year t is proportional to $n_{ta}f(a)$, where n_{ta} counts available papers of age a. A conditional target likelihood compares discrete-grid lognormal, exponential, power-law and uniform kernels using AIC on identical observations and support. We repeat the comparison by publication cohort (1970–1989, 1990–1999, 2000–2009), accounting for each cohort's actual exposure. This estimates marginal ageing; degree, fitness and venue coverage can also influence this curve.

M2c-3: densification and diameter. We fit $\log E(t) = a \log N(t) + b$ over 1990–2020 and repeat over split periods and an earlier observation cutoff. Year-resampling intervals are descriptive because cumulative observations are dependent. Effective diameter is the 90th percentile of weak shortest-path distances within the largest connected component, using 128 uniformly sampled source nodes per year. Sensitivity checks use 64 and 256 sources. We save the component size and number of sources with every estimate [18].

M2c-4: citation dynamics and prediction. We fit the Wang–Song–Barabási curve [37],

$$c(t) = m\left[\exp\left\{\lambda\Phi\left(\frac{\log t - \mu}{\sigma}\right)\right\} - 1\right], \qquad (3)$$

where m is fixed to the mean observed within-subgraph reference count of papers published through 1989. Bounded multistart least squares minimizes squared $\log(1+c)$ errors; parameter bounds and boundary flags are preserved. For descriptive full-history fits we randomly select at most 500 papers born in 1990–2005 with at least 10 citations by age 15 and fit years 1–15. The resulting collapse is in-sample evidence only. For prediction, selection uses only the first five years: at least five observed citations by age five, followed by a fixed random sample of at most 500 papers. Using only years 1–5, we compare WSB against no-ageing PA, $c(t) = m[\exp(rt) - 1]$, and linear extrapolation, $c(5)t/5$. The primary task predicts cumulative citations at age 15, ten years after the five-year training cutoff. Every selected paper (born 1990–2005) has its age-15 target within the 2020 observation window. A secondary task predicts each paper's observed 2020 total, with a variable forecast horizon. Neither target represents lifetime citations; within-venue coverage remains a limitation. This fixed-horizon adaptation differs from the original paper's Figure 4, which trains on 5 or 10 years and evaluates at age 30 [37]. We report MAE, median absolute error, RMSLE, paired bootstrap differences (1000 resamples) and an interior-fit sensitivity check. Boundary fits remain in the primary evaluation so that difficult cases are not silently discarded. Boundary status and optimizer success are recorded separately; a constrained optimum is not, by itself, a failure to converge.

P3: citation-target randomization. Twenty realizations per model preserve each paper's publication year and exact number of distinct observed references, with no replacement and strictly older targets. P3a selects eligible targets uniformly. P3b weights targets by a nonparametric age kernel fitted by conditional target likelihood, accounting for each citing year's available age distribution, and removes degree preference. Its age weights are calibrated from the data, so agreement on ageing is not an independent confirmation. We report null means and sample standard deviations; empirical tail probabilities have minimum resolution $1/21$. Both models fix $N(t)$ and $E(t)$ and hence exactly preserve the densification exponent. They can test diameter and target preference, but cannot supply a significance test for densification itself.

Reproducibility. All significant CPU workloads run on Vanda compute nodes through PBS without GPU allocation. Fixed seeds, input/source checksums, environment

lock file, filtering counts, individual forecasts and every randomized edge list are saved. Robustness checks cover time splits, cutoff 2015, degree offsets, same-year edges, core database venues, diameter sampling, and 10 versus 20 null realizations.

M4 cuts the event sequence into consecutive windows of width Δt and aggregates each into a weighted undirected snapshot (weight = number of events). Windows with very few events are dropped (fewer than 100 for PS, which removes nights; fewer than 50 for CM), and only nodes active in a window belong to its snapshot. The main settings are $\Delta t = 15\,\text{min}$ for PS and 7 days for CM. Every snapshot and the static aggregate are partitioned with Leiden [36] maximising modularity with resolution γ [29]. A multislice partition [23] with interslice coupling $\omega = 1$ gives labels that are consistent across windows; we use it only for tracking, because the coupling itself pulls consecutive partitions together.

Partitions are compared with NMI [4], always on the nodes the two partitions share. For each window we record the NMI with the previous window (only for back-to-back windows, not across a night), with the static partition, and for PS with the classes (teachers excluded). As a noise floor, every window is partitioned a second time with another random seed. All measures are recomputed on 20 P1 samples: P1 keeps the static graph, and hence the static partition, and the number of events per window, so H5 predicts that real snapshots agree less with each other and with the static partition than P1 snapshots do. We repeat the analysis for $\gamma \in \{0.5, 1, 2\}$ and $\Delta t \in \{5, 15, 60\}$ min (PS) or $\{1, 7, 30\}$ days (CM).

4.5. Paths, centrality and spreading (M3)

M3a averages earliest-arrival reachability over 16 PS clock-hour starts and 20 CM starts spread over the first 90% of the selected interval. Paths use strictly increasing timestamps; simultaneous events cannot form a chain. Temporal reachability and static reachability use the same period node universe. The ratio R uses the full selected period's static graph; late starts therefore include a finite-horizon boundary effect. Latency is conditional on reachability.

M3b compares static and temporal closeness. Static closeness is harmonic closeness [20], $C^{\text{s}}(i) = \sum_{j \neq i} 1/d(i,j)$, on the aggregate (outgoing distances for CM), with hop distance (unweighted) or distance $1/w_{ij}$ (weighted). Temporal closeness [15, 25] uses the same time-respecting paths as M3a: from a start time t_0, an event (u, v, t) can be used only if u was reached strictly before t, and v is then reached at t, so no path takes two hops within one timestamp. With $a_{t_0}(i \to j)$ the earliest arrival time,

$$C^{\text{t}}(i) = \frac{1}{|\mathcal{T}|} \sum_{t_0 \in \mathcal{T}} \frac{1}{N-1} \sum_{j \neq i} \frac{1}{a_{t_0}(i \to j) - t_0}, \quad (4)$$

where unreachable nodes contribute zero. $\mathcal{T}$ holds every full hour from 09:00 to 16:00 on both PS days (16 start times) and 20 evenly spaced times over the first 90% of the CM period. Agreement is measured by Kendall's τ_b and the overlap of the top 10% nodes. P1 leaves the static graph and so C^{s} unchanged; recomputing C^{t} on 20 P1 samples therefore isolates the effect of timing.

M3c uses 200 event-started seeds per network and four parameter combinations. PS uses $(\beta, D) = (.05, 4\,h), (.02, 4\,h), (.10, 4\,h), (.05, 1\,h)$; CM uses $(.20, 14\,d), (.05, 14\,d), (.50, 14\,d), (.20, 3\,d)$. The first setting is primary. Event positions are drawn uniformly from the first 70% of each stream. The source is the sender in CM and a random endpoint in PS. Sources are infected at the selected event time; only later events transmit. We report final size, probability of reaching 5% of nodes, and peak-delay ratios conditional on major outbreaks. There are 20 P1 and 20 P2 networks in each primary run. Simultaneous infections and recoveries are combined before measuring prevalence.

4.6. Randomized reference models

No single temporal counterpart of the configuration model exists, so one switches off selected correlations with different models and compares the outcomes [6, 12].

For PS and CM, we consider two complementary types of randomization, corresponding to *randomly permuted times* (RP) and *randomized edges* (RE) in the naming of Holme and Saramäki [12].

P1 (RP): Timestamp permutation. P1 randomly permutes timestamps among events while keeping their endpoints unchanged. It preserves the static graph, the number of contacts of every pair, and the global timestamp distribution, while disrupting the original temporal correlations. For H2 and H3, we generate 20 realizations and report the null mean and standard deviation.

P2: Structural randomization. We distinguish two variants of edge randomization.

P2a (RE, original proposal). The original method rewires the static graph in a degree-preserving way while each pair's contact sequence moves with its link. It aims to destroy the wiring diagram while preserving per-link temporal activity patterns and the overall event rate. This variant is retained as the original methodological proposal but was not evaluated in the Member C experiments.

P2b (constrained endpoint swaps, implemented). Our implementation randomly swaps endpoints between event pairs while keeping timestamps fixed. For undirected PS, endpoint orientations are randomized before swapping; for directed CM, sender and receiver roles are preserved. Swaps creating self-loops or duplicate edge-time events are rejected.

P2b preserves the total event count, timestamp multiset,

and node event participation counts (separate sender and receiver counts for CM). Unlike P2a, it does not necessarily preserve static simple-graph degrees or individual edge contact sequences.

We attempt five swaps per event and validate the structural invariants using a fixed random seed of 42. These checks do not establish uniform sampling or sufficient mixing of the randomization process.

Comparing temporal and structural randomization helps distinguish the effects of event timing from network topology. Combined models (RE+RP) and fully random times (RT) remain possible extensions.

4.7. Measuring the effect of time

Raw metrics cannot be compared across a contact network, a message network and a citation network, so we define one normalised measure per hypothesis: H1 window sensitivity (relative change of a metric over the Δt grid); H2 differences in pooled burstiness and memory relative to P1, $B_{\text{real}} - \overline{B}_{\text{P1}}$ and $M_{\text{real}} - \overline{M}_{\text{P1}}$; H3 differences in adjacent-window link Jaccard relative to P1, $J_{\text{real}} - \overline{J}_{\text{P1}}$; H4 exponents α and a; H5 mean consecutive-snapshot NMI and dynamic-vs-static NMI; H6 reachability ratio R and Kendall τ(static, temporal); H7 slowdown $T_{\text{real}}/T_{\text{null}}$, where T is the time to the epidemic peak.

4.8. Implementation

All code is in Python. We use NetworkX for static graphs, a strict-time all-source event-scan algorithm for earliest arrivals, NumPy with optional Numba acceleration for SIR, and igraph/Leiden for per-snapshot and multislice partitions. Null models, the attachment-kernel estimator and the SIR process are custom code, unit-tested on hand-built examples whose answers are known (reachability, SIR, preserved quantities of P1 and P2). Random seeds are fixed and every figure is generated by one script.

Supplementary robustness protocol. PS/CM use elapsed-time midpoint splits; PS starts retain their clock-hour convention, while CM starts are recalculated within each half. Null sensitivity compares the first 20 with all 60 seeded networks, using the same real network. D retains four SIR parameter settings. E repeats its primary window/resolution on the halves and 20/60 nulls; its existing nine-setting resolution grid is retained. CIT uses the 1990–2020 analysis interval split into 1990–2004 and 2005–2020, retaining earlier papers as eligible targets. PA and age statistics receive additional P3 draws; existing densification/diameter checks remain separate, and WSB's fixed-horizon forecast test is not interpreted as a stationary half-window experiment.

5. Results

The results follow the four method groups. All primary PS/CM cells now contain measurements; null comparisons use 20 networks per family. The supplementary robustness tables retain individual networks and elapsed-time half-window results.

5.1. Aggregation and time windows

Contact and messaging networks. We use nonoverlapping windows anchored at the first observed event, with all nodes in the selected period retained in every window. Empty windows and the final partial window are included; overnight inactivity in PS therefore contributes to the average. CM clustering is measured on the undirected projection; density and total degree retain message direction. The grids span 5 minutes–1 day (PS) and 1 hour–30 days (CM). For each metric, sensitivity is the range of its window-averaged values divided by their mean across grid widths.

Metric sensitivity	PS	CM
Density	3.359	3.862
Mean total degree	3.359	3.862
Clustering	1.628	3.909
Weak components	1.696	0.364
Weak LCC fraction	2.066	2.912

Table 3. Measured relative ranges over the stated grids. These are descriptive sensitivity measures, not temporal-null significance tests.

Citation network (CIT). In 2020, annual and cumulative graphs on the same 19,150 born nodes contain 1,450 and 73,269 citations. Mean total degree is 0.151 versus 7.652, weak largest-component fraction 0.052 versus 0.641, and cumulative weak clustering 0.124. Annual clustering is identically zero: strict older-target citations make each annual snapshot bipartite. This is a consequence of the representation, not independent evidence for temporal correlations. Over 1-, 2-, 5- and 10-year windows, relative ranges average 1.798 for density/degree, 2.361 for clustering and 1.114 for largest-component fraction. **CIT verdict on H1:** window choice changes structural descriptions; some differences are definitional. The node denominator and year convention must accompany every comparison.

Window interpretation. Figure 4(c,d) shows that longer windows connect a larger fraction of born papers and admit cross-year triangles. These effects require a sequence of snapshots; one final aggregate cannot reveal when connectivity appears. The relative range is a window-sensitivity

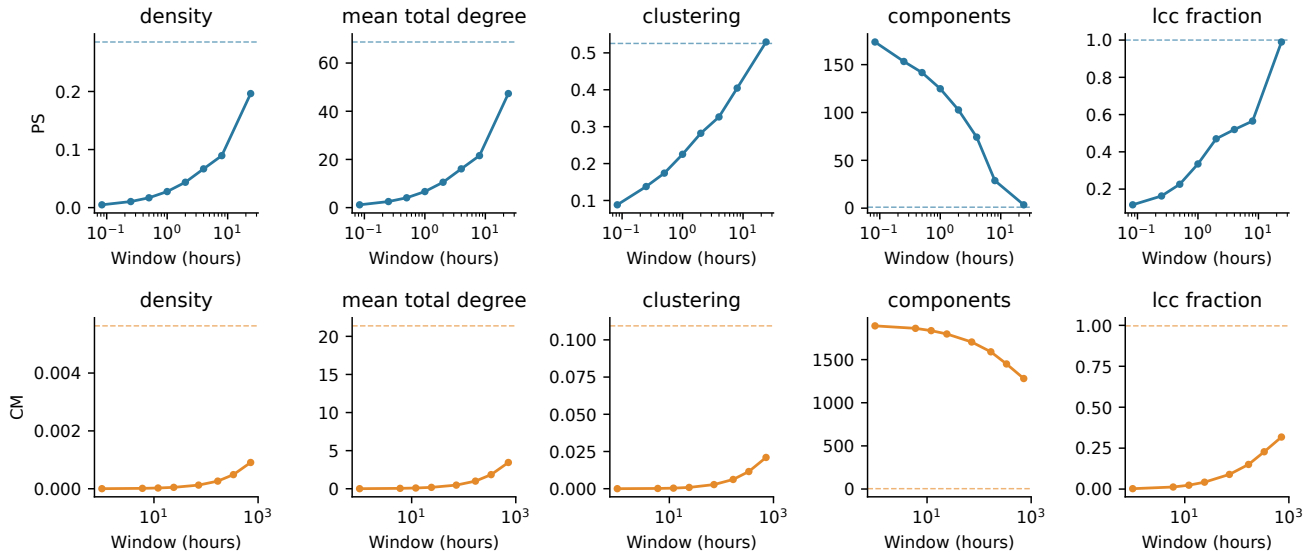


Figure 3. Structural metrics averaged over windows (PS top, CM bottom; separate scales). Dashed lines show full aggregates. The denominator includes inactive nodes.

statistic, not a significance test against a temporal null, and identical density/degree sensitivity follows from the common denominator.

Verdict on H1. Supported as a representation effect in all three datasets. Numerical sensitivity depends on the width grid, alignment, inclusion of inactive nodes and observation span; these values alone do not establish a universal ranking of network types.

5.2. Activity patterns

Inter-event intervals are pooled within dyads. In PS, contiguous 20-second contacts are merged into episode onsets before burstiness, memory and repeated-tie analysis. Memory pools adjacent interval pairs within each tie, without joining different ties. Lifetimes and Jaccard use the original cleaned records. P1 uses seeds 0–19; coincident events produced by timestamp shuffling are retained to preserve message counts. Canonical input ordering fixes each seeded realization; null values below are recomputed rather than copied from earlier rounded report tables. In PS, $\Delta B = +0.0163$ and $\Delta M = -0.0511$. Mean tie lifetime is 0.444 days, versus 0.690 days under P1. In CM, $\Delta B = +0.3360$ and $\Delta M = -0.1284$. Mean tie lifetime is 4.756 days, versus 30.028 days under P1.

Verdict on H2 and H3. Burstiness exceeds P1 in both datasets, while pooled positive memory is weaker than P1. Thus H2 has a mixed verdict. H3 depends on the statistic: real ties are temporally concentrated, and adjacent overlap is below P1 in PS but above it in CM at the primary widths.

Network	Metric	Real	P1 mean ± SD
PS	B	0.535	0.5184 ± 0.0004
PS	M	0.020	0.0710 ± 0.0016
PS	Repeated ties	0.625	0.6568 ± 0.0001
PS	Jaccard	0.220	0.3668 ± 0.0010
CM	B	0.581	0.2451 ± 0.0012
CM	M	0.100	0.2288 ± 0.0057
CM	Repeated ties	0.495	0.4949 ± 0.0000
CM	Jaccard	0.096	0.0690 ± 0.0009

Table 4. Measured activity and persistence. Jaccard uses 1-hour PS and 7-day CM windows, including empty windows when their union is nonempty.

The PS lifetime is recomputed over canonical undirected dyads; the earlier report's lifetime value is superseded by this shared-interface calculation. Pooling across heterogeneous ties and short observation windows limit generalization.

5.3. Structural evolution

Preferential attachment (H4a). The conditional kernel $(k+1)^{\alpha}$ gives $\alpha = 0.714$, with a descriptive 95% year-bootstrap interval $[0.693, 0.730]$. Twenty P3a samples yield 0.0006 ± 0.0051 and age-only P3b yields -0.6512 ± 0.0053. The observed exponent exceeds all null values (empirical one-sided probability $1/21$ per comparison, with coarse resolution). The exponent stays positive under time splits, offsets 0.5/2 and a core venue-family subset. **Verdict:** sublinear degree preference is supported within this corpus, beyond the age-only reference; this does not identify a causal

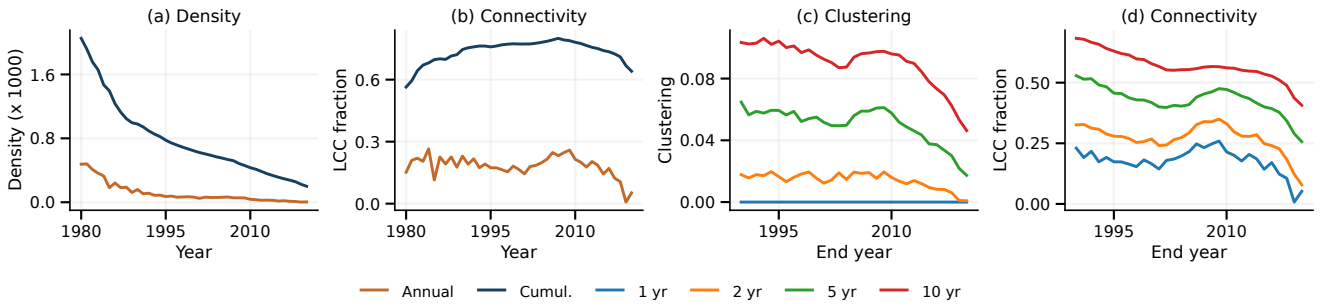


Figure 4. CIT representation effects. (a,b) Annual versus cumulative density and weak-LCC fraction; (c,d) clustering and connectivity for 1-, 2-, 5- and 10-year trailing windows. All fractions include born isolates. Annual zero clustering follows from strict publication-year ordering; it is not separate mechanism evidence.

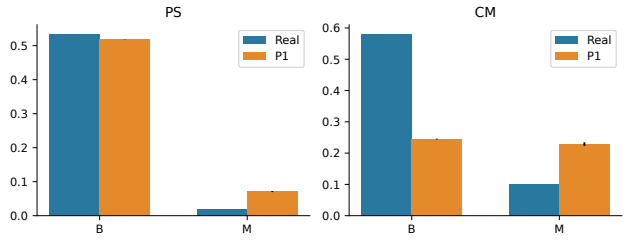


Figure 5. Pooled burstiness and memory relative to 20 P1 samples.

PA mechanism.

Ageing (H4b). Lognormal target preference is preferred to exponential, power-law and uniform kernels, with respective ΔAIC 537.8, 9,360.5 and 41,796.7. The lognormal parameters are $(\mu, \sigma) = (1.858, 1.108)$; its underlying scale e^{μ} is 6.41 years. Mean age over all retained citations is 5.711 years, versus 11.457 ± 0.034 for P3a and 5.717 ± 0.018 for calibrated P3b. Lognormal is preferred in all three target cohorts, although recent-cohort tails remain censored. **Verdict:** age-independent choice is inadequate; P3b's agreement is calibration, not independent confirmation of ageing.

Densification and diameter (H4c). For 1990–2020, $a = 1.343$ ($R^2 = 0.991$), with descriptive year-resampling interval $[1.288, 1.406]$. Growth is weaker in 2005–2020 ($a = 1.065$) than 1990–2004 (1.455); cutoff 2015 gives 1.427, inclusion of year ties 1.324, and the core venue subset 1.410. Densification is superlinear edge growth, not increasing directed density. The sampled weak-LCC effective diameter is 6 hops in every year; 64/256-source endpoint checks agree. Isolates lie outside the weak LCC and cannot directly explain this constant within-component statistic. P3 fixes $N(t)$ and $E(t)$, hence also a. **Verdict:** densification is observed, but diameter shrinkage is not detected at the sampled 90th-percentile resolution.

Citation dynamics and prediction (H4d). For 500 full-history fits, median WSB parameters are $(\lambda, \mu, \sigma) = (2.717, 0.834, 1.238)$, median RMSLE is 0.074 and scaled-collapse RMSE is 0.022. These are fitted-sample diagnostics. On 500 papers selected using only their first five years, the primary ten-year-ahead task predicts cumulative citations at age 15. Its MAE is 15.144 for WSB, 17.064 for linear extrapolation and 27,588 for no-ageing PA; at the observed 2020 endpoint it is 28.016, 27.271 and 2.79×10^6. The paired MAE differences WSB minus linear have 95% intervals $[-4.471, 0.806]$ and $[-4.764, 8.128]$, both including zero. Forecast fits hit parameter bounds for 193/500 papers (38.6%), while all 500 optimizations report success. Boundary status signals short-history identifiability or bound sensitivity, not optimizer non-convergence; all remain in primary scoring. **Verdict:** compact descriptive fits are supported, but a clear predictive advantage over the simple linear baseline is not established. All age-15 targets are observed; the secondary 2020 target has a variable horizon. Both remain within-corpus counts, not lifetime impact.

Dynamic vs. static communities. In PS (Fig. 8) the snapshot communities mostly match the classes outside the midday break (NMI $\approx$ 0.85–0.95), as does the static partition (six communities, NMI 0.86 with the classes). Between about 12:00 and 14:00 on both days the NMI with the classes falls to ≈ 0.5, and the NMI with the static partition falls just as far. Under P1 both curves stay flat (class NMI 0.86 ± 0.002), so the drop comes from when contacts happen, not from who is in contact. It is not algorithmic noise either: two Leiden runs on the same window agree with NMI 0.98 on average. Averaged over the day, real snapshots agree with the static partition less than P1 snapshots (0.72 vs. 0.94 ± 0.003).

CM behaves differently (Fig. 9). Weekly snapshots agree only weakly with each other and with the static partition (NMI 0.52 and 0.34), and P1 gives almost the same values (0.42 and 0.32). The snapshots are sparse (326 of 1,899

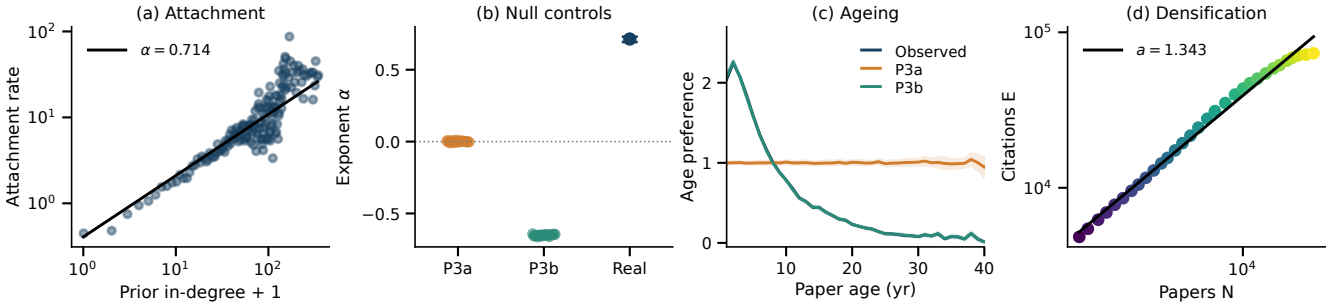


Figure 6. CIT growth mechanisms. (a) Prior-degree kernel (displayed classes have at least five hits; the exponent uses all supported classes). (b) Twenty draws per P3 family and the observed exponent with its descriptive year-bootstrap interval. P3b controls age preference while removing explicit degree preference. (c) Exposure-adjusted age preference, null mean ± SD; P3b agreement is calibration. (d) Cumulative edge growth; colour progresses from 1990 to 2020. The constant sampled diameter is reported in text rather than as a flat curve.

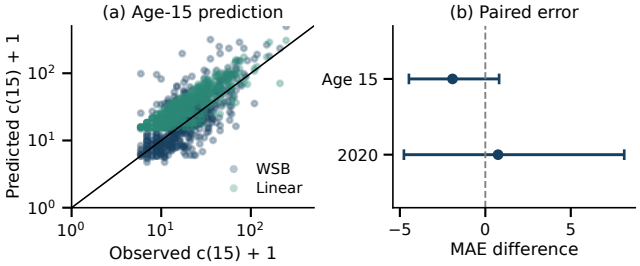


Figure 7. WSB held-out prediction. (a) Age-15 cumulative citations predicted from ages 1–5; diagonal denotes exact prediction. (b) Paired-bootstrap 95% intervals for MAE(WSB)−MAE(linear), using all 500 papers. Both cross zero. Age 15 is the primary fixed-horizon task; 2020 is a secondary variable horizon.

users active per week on average), and during the busy first two months two runs of the same algorithm agree only at NMI ≈ 0.65–0.7. Where real and null differ, real snapshots are *more* similar to their predecessor than random ones, which points to persistent conversation groups rather than reorganisation. The same contrast appears at the level of ties (Sec. 5.2): the mean Jaccard overlap of consecutive edge sets is lower than under P1 in PS (0.22 vs. 0.37, 1-h windows) but higher in CM (0.096 vs. 0.069, 7-day windows).

One property of P1 matters for reading these numbers. Permuting timestamps spreads every node's events over the whole observation period, so a P1 window contains almost all nodes (240 vs. 175 active nodes per 15-min PS window) and resembles a sample of the static graph. Part of the gap between real and P1 agreement with the static partition therefore reflects *who is active when*, not only *who meets whom*. The lunch-time drop in class NMI is not affected by this, since it compares real snapshots with external labels.

Figure 10 repeats the comparison for three window widths and three resolutions. In PS, real snapshots agree less with the static partition than P1 in all nine settings (difference −0.03 to −0.23). In CM the difference stays within ±0.06, and within ±0.025 except for 30 days with $\gamma = 0.5$. The comparison with the previous window is not robust in PS: at 15 and 60 min real snapshots change more than P1 ones, but at 5 min the order flips, because a 5-min P1 snapshot is a small random sample of the static graph and is itself noisy. Larger γ gives more, smaller communities without changing any of these signs.

Verdict on H5. *Supported for face-to-face contacts, not supported for online messages.* In PS the static partition recovers the classes but hides the daily mixing at lunch, which P1 does not reproduce. In CM the static partition summarises weekly snapshots as well as it summarises time-shuffled ones, and snapshots are, if anything, more persistent than random.

5.4. Paths, centrality and spreading

Temporal reachability (M3a). We average over 16 clock-hour starts in PS and 20 evenly spaced starts in the first 90% of the CM interval. Paths require strictly increasing timestamps; unreachable pairs contribute zero to reachability. Mean latency is averaged over reachable pairs at each start and then over starts. R divides temporal reachability by static reachability on the full selected period, with the same nodes and direction.

Interpretation. Both networks lose reachable pairs when observation starts at different times. The reduction is stronger in CM, where direction and late starts restrict causal paths. P1 preserves the static graph; the real–P1 difference isolates timing conditional on that graph. P2 also changes topology and cannot isolate timing alone.

Centrality. Static and temporal closeness rankings are clearly correlated in both networks (Tab. 6 and Fig. 12;

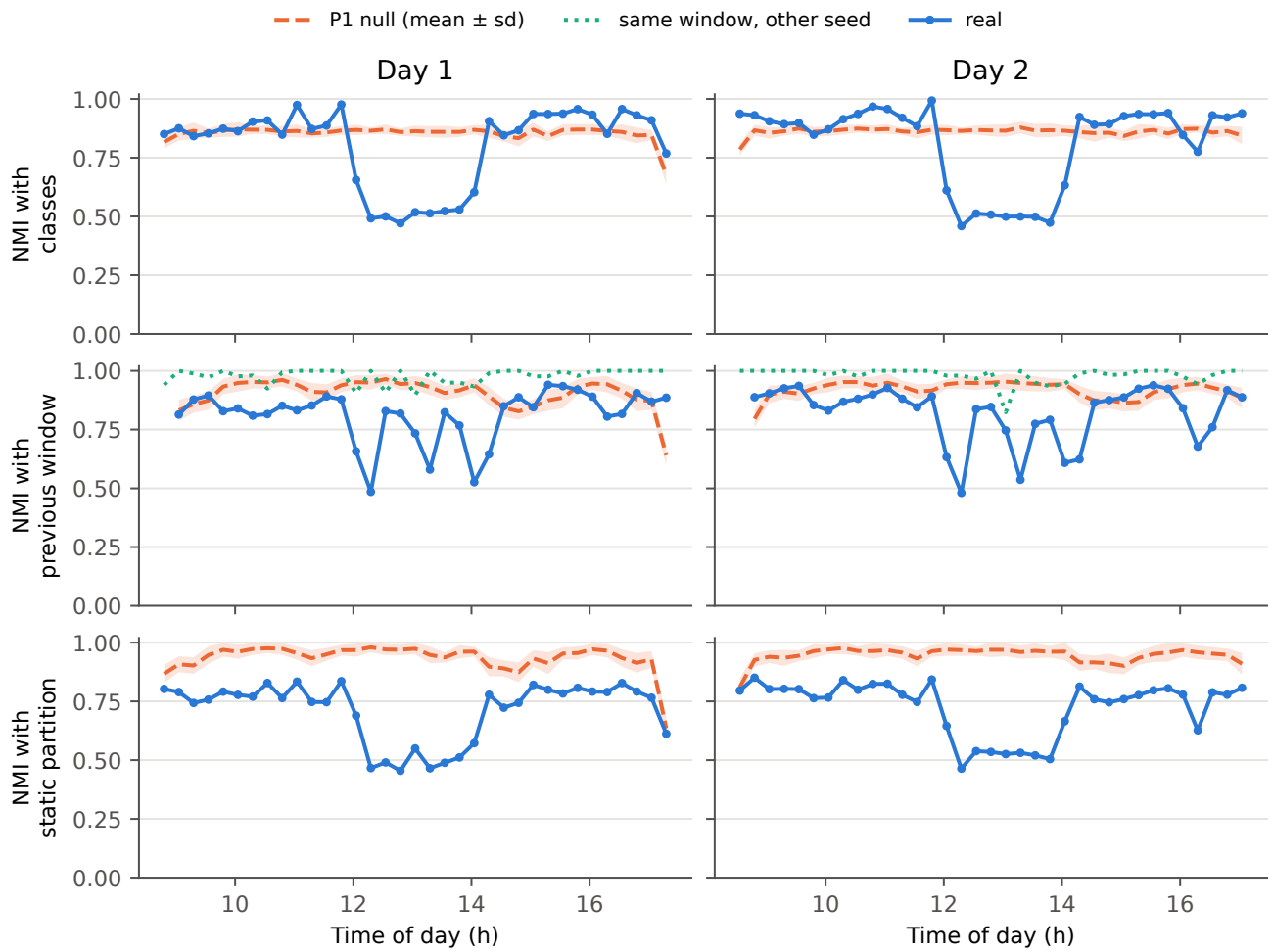


Figure 8. Snapshot communities in PS (15-min windows, $\gamma = 1$). Rows: NMI of each snapshot partition with the classes, with the previous window and with the static partition. Blue: real; orange: P1 (mean ± sd, 20 samples); green dotted: same window partitioned with another seed (noise floor).

Dataset	Metric	Real	P1	P2
PS	Reach	0.866	1.000	1.000
PS	R	0.866	1.000	1.000
PS	Delay (h)	3.390	0.152	0.124
CM	Reach	0.140	0.223	0.165
CM	R	0.205	0.326	0.237
CM	Delay (h)	948.993	906.718	912.285

Table 5. Multi-start averages; P1/P2 columns average 20 independent networks. Individual starts and null SDs are retained in the result tables.

$\tau = 0.71$ in PS and 0.78 in CM, hop distance), so the static network shows roughly which nodes are central. It is much less reliable about which nodes are *most* central: only half of the PS top 10% and two thirds of the CM top 10% co-

	Measure	Real	P1
PS	τ, unweighted	0.71	0.55 ± 0.01
	τ, weighted	0.39	0.69 ± 0.01
	top-10%, unweighted	0.50	0.51 ± 0.07
	top-10%, weighted	0.29	0.60 ± 0.06
CM	τ, unweighted	0.78	0.85 ± 0.00
	τ, weighted	0.74	0.86 ± 0.00
	top-10%, unweighted	0.65	0.74 ± 0.01
	top-10%, weighted	0.57	0.72 ± 0.02

Table 6. Agreement of static and temporal closeness rankings: Kendall τ_b and overlap of the top 10% nodes. P1: temporal closeness on 20 samples (mean ± sd); the static ranking is unchanged by P1.

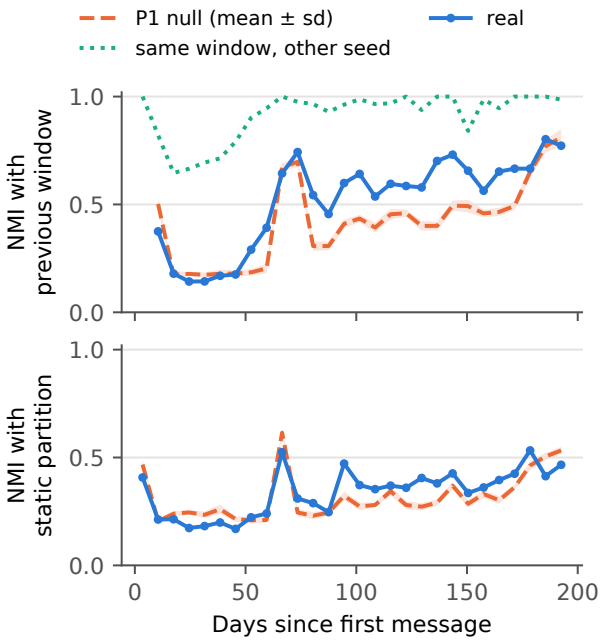


Figure 9. Snapshot communities in CM (7-day windows, $\gamma = 1$); encoding as in Fig. 8. CM has no ground-truth labels.

incide. In CM the real agreement is somewhat lower than under P1 (0.78 vs. 0.85), so event order makes static closeness a slightly worse guide. The CM τ is also helped by the 549 users (29%) who never send a message and have zero closeness in both rankings (the tied block in Fig. 12).

In PS the effect depends on edge weights. Weighted static closeness agrees poorly with temporal closeness ($\tau = 0.39$, top-10% overlap 0.29), whereas after P1 the agreement is high (0.69 ± 0.01). In the school, most contact weight comes from repeated contacts within a class during lessons; these happen while the class is already connected and hardly shorten time-respecting paths to the rest of the school. Under P1 the same contacts are spread over the day, and a heavy edge becomes a frequent chance to transmit. Without weights the effect reverses (real 0.71 vs. P1 0.55): reaching the school quickly depends on having many distinct partners, which in PS largely means mixing at breaks.

Rankings part of H6: partly supported. The rankings are correlated ($\tau = 0.71$–0.78) but the top nodes differ by 35–50%, and in face-to-face data weighting edges by contact frequency makes the static ranking misleading ($\tau = 0.39$).

Event-driven SIR (M3c). Each network uses 200 seeds at randomly selected events in its first 70% of event positions. CM seeds use the sender; PS randomly selects either endpoint. The selected seed event itself is excluded by the strict-time convention. Infectious duration is fixed, transmission probability is per record, and recovered nodes remain immune. Final size includes the seed. A major outbreak reaches at least 5% of period nodes. Peak delays are averaged over major outbreaks only; $T_{\mathrm{real}}/\overline{T}_{\mathrm{null}}$ is a conditional ratio of means. Nulls with no major outbreaks give an undefined peak, not a zero delay. Simultaneous infections and recoveries are grouped before measuring prevalence; the earliest maximum is selected. The revised design produces nontrivial CM outbreaks. Absence of secondary infections in the earlier run arose from infecting seeds at the observation boundary with an unsuitable short duration; it is not evidence for an intrinsic property of CM.

Verdict on H6 and H7. H6's reachability component is supported under the multi-start protocol in both datasets; rankings show a partial effect, with correlated rankings but different top nodes. H7 is conditional on parameters and the reference model. PS peaks later under its primary setting; CM peaks earlier than P1 across the four tested full-period settings and is close to P2. The claim that slowing is strongest in CM is therefore not supported. Peak timing among major outbreaks does not by itself measure population-wide epidemic speed.

5.5. CIT model diagnostics and robustness

Null controls and ageing alternatives. Figure 6(b) shows the observed exponent and both null ensembles; Tab. 8 reports the age-kernel comparison without a duplicate AIC plot. P3b matches mean citation age because its age weights were fitted to these data; that agreement is a property of the control, not independent evidence for ageing. With only 20 realizations per null, an observed value above all nulls has empirical one-sided probability $1/21$; this does not justify a precise tail estimate. The ageing comparison is on identical risk sets and observations.

Fit quality does not establish predictability. The full-history sample is 500 of 1,019 papers with at least ten citations by age 15; the forecast sample is 500 of 1,220 eligible papers selected only by having at least five citations in their first five years. Both samples are drawn with a fixed seed. Median full-history RMSLE is 0.0736 and median scaled-collapse RMSE 0.0220 (90th percentile 0.0453); 10.2% of full-history fits hit a parameter bound. These are in-sample diagnostics. Held-out scoring in Tab. 9 and Fig. 7 provides the separate predictive test. All 500 short-history optimizations report success, while 193 touch a bound; these are distinct diagnostics. In particular, a collapse obtained using parameters fitted to the same curve is not independent evidence for universal citation dynamics.

Sensitivity of the conclusions. Table 10 reports time splits, cutoff and corpus-selection checks. The sign of degree preference and superlinear full-period growth persist,

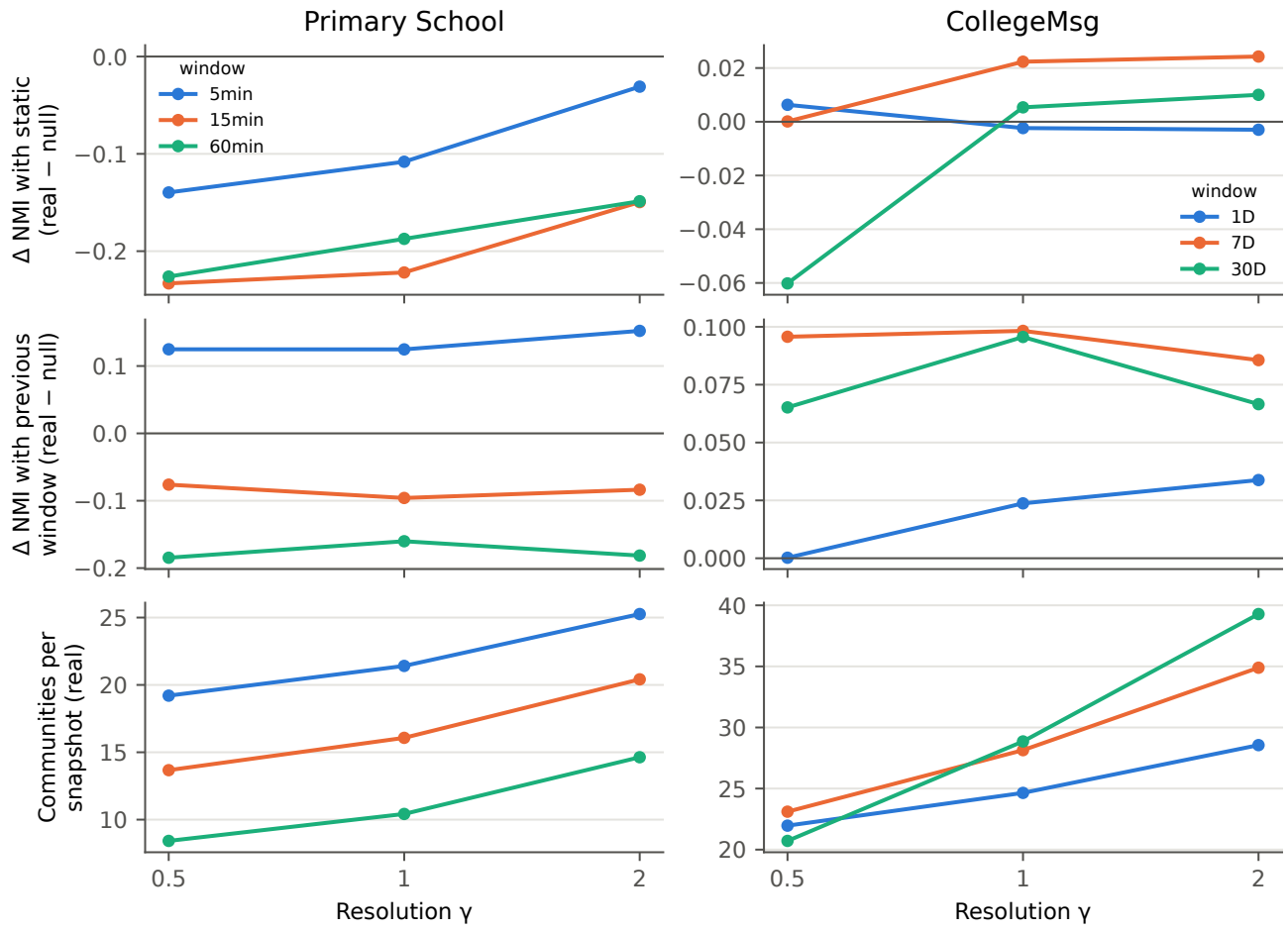


Figure 10. Sensitivity of H5 to window width and resolution γ. Top two rows: real minus P1 mean of the window-averaged NMI with the static partition and with the previous window (negative: real snapshots agree less). Bottom: communities per real snapshot.

		Final infected fraction			Major outbreak probability			Peak ratio	
Dataset	(β, D)	Real	P1	P2	Real	P1	P2	vs P1	vs P2
PS	(0.05, 4 h)	0.629	0.776	0.823	0.860	0.908	0.920	1.160	1.214
PS	(0.02, 4 h)	0.373	0.420	0.559	0.700	0.754	0.778	1.000	1.088
PS	(0.1, 4 h)	0.734	0.893	0.918	0.910	0.960	0.973	1.311	1.340
PS	(0.05, 1 h)	0.373	0.634	0.674	0.745	0.844	0.852	1.021	1.265
CM	(0.2, 336 h)	0.256	0.250	0.276	0.825	0.791	0.799	0.888	0.995
CM	(0.05, 336 h)	0.026	0.013	0.025	0.215	0.082	0.192	0.934	0.991
CM	(0.5, 336 h)	0.460	0.506	0.530	0.975	0.954	0.965	0.924	0.982
CM	(0.2, 72 h)	0.074	0.043	0.066	0.555	0.342	0.428	0.720	1.013

Table 7. Four SIR settings per dataset, 200 seeds per network, 20 nulls per family. The first setting in each dataset is primary. The peak-ratio population is restricted to major outbreaks and can differ between real and null networks.

but a is substantially weaker in the late period. Degree offsets 0.5 and 2 give $\alpha = 0.604$ and 0.858, so the numerical exponent is convention-dependent. Including 5,486 year ties gives $a = 1.324$; ties remain excluded from the primary DAG. Effective-diameter endpoint estimates remain 6 hops with 64, 128 or 256 sources. First-ten versus all-twenty null means of α are 0.00055/0.00059 (P3a) and $-0.65320/-0.65118$ (P3b). Lognormal ageing is preferred in all target cohorts and both observation halves, but the recent cohort's fitted tail extends beyond its observed follow-

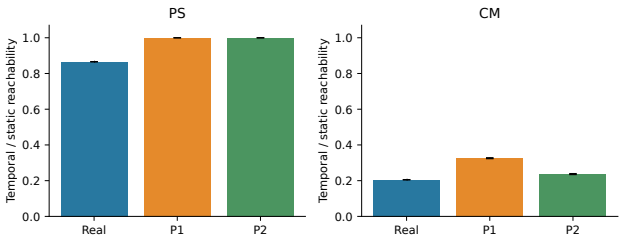


Figure 11. Temporal/static reachability ratios; error bars are the null-network SD.

Age kernel	Parameters	ΔAIC
Lognormal	2	0.0
Exponential	1	537.8
Power-law	1	9,360.5
Uniform	0	41,796.7

Table 8. CIT age-preference alternatives on common observed support. AIC selects among the declared models, not among every possible ageing mechanism.

Target	Model	MAE	RMSLE
15 yr	WSB	15.144	0.601
	Linear	17.064	0.662
	No-age PA	27,588	4.601
2020	WSB	28.016	0.778
	Linear	27.271	0.876
	No-age PA	2.79×10^{6}	7.000

Table 9. Matched forecasts on 500 selected papers using only years 1–5 for training. Age 15 is fully observed for the selected cohort and is the primary fixed ten-year-ahead task; 2020 is a secondary, variable-horizon within-corpus total. The exponential no-age PA baseline over-extrapolates; beating it alone is a weak test. The paired MAE intervals against linear both include zero.

Variant	α	a
1990–2020 (primary)	0.714	1.343
1990–2004	0.684	1.455
2005–2020	0.721	1.065
Cutoff 2015	0.722	1.427
Core venue families	0.731	1.410
Same-year edges included	–	1.324

Table 10. CIT robustness from the saved run. Core venue families contain 14,235 nodes and 47,009 primary edges; the year-tie variant is structural only.

up. These checks support qualitative stability without eliminating coverage bias, right censoring or unobserved fitness.

	Measure	Full	First	Second	60 nulls
PS	H1 density	3.359	2.596	2.487	–
CM	H1 density	3.862	3.623	3.478	–
PS	H2 ΔB	0.016	0.116	0.120	0.016
CM	H2 ΔB	0.336	0.381	0.313	0.336
PS	H6 R	0.866	0.815	0.765	0.866
CM	H6 R	0.205	0.314	0.177	0.205
PS	H7 ratio	1.160	1.081	1.095	1.161
CM	H7 ratio	0.888	0.856	0.865	0.884
PS	H5 NMI	0.816	0.812	0.818	0.816
CM	H5 NMI	0.519	0.374	0.663	0.519
CIT	H4 α	0.714	0.684	0.721	0.714

Table 11. Elapsed-time halves for PS/CM and analysis-year halves for CIT. A uses no null. The 60-null run nests the first 20; H7 ratios condition on major outbreaks. A and E real values do not depend on null count, while H2 differences and H7 ratios do.

5.6. Observation and null-count sensitivity

Primary H7 ratios remain on the same side of one under the 20/60 comparison: PS changes from 1.160 to 1.161 and CM from 0.888 to 0.884. Both primary half-period ratios preserve those directions. This does not hold uniformly over parameter settings: at PS $\beta = .02$, $D = 4$ h the full-period ratio moves from 1.0005 (20 nulls) to 0.9992 (60 nulls), so no directional slowing claim is supported there. PS's one-hour setting peaks earlier than P1 in the first half (ratio 0.874) despite a full-period ratio above one. In CM's second half, two settings produce no major real outbreaks and their conditional peak ratios are undefined; these are not treated as zero. Individual realizations, sample SDs and 20/60 changes are retained in the supplementary CSV files. Splitting changes node populations and the time available for paths and outbreaks, so agreement of signs is more informative than equality of raw effect sizes. The established E resolution grid is retained. CIT retains its original history for PA exposure and eligible targets; its diameter is measured on cumulative endpoint graphs. WSB uses the original age-15 forecast test, additionally stratified by early and late publication cohorts, rather than being treated as a stationary half-window test.

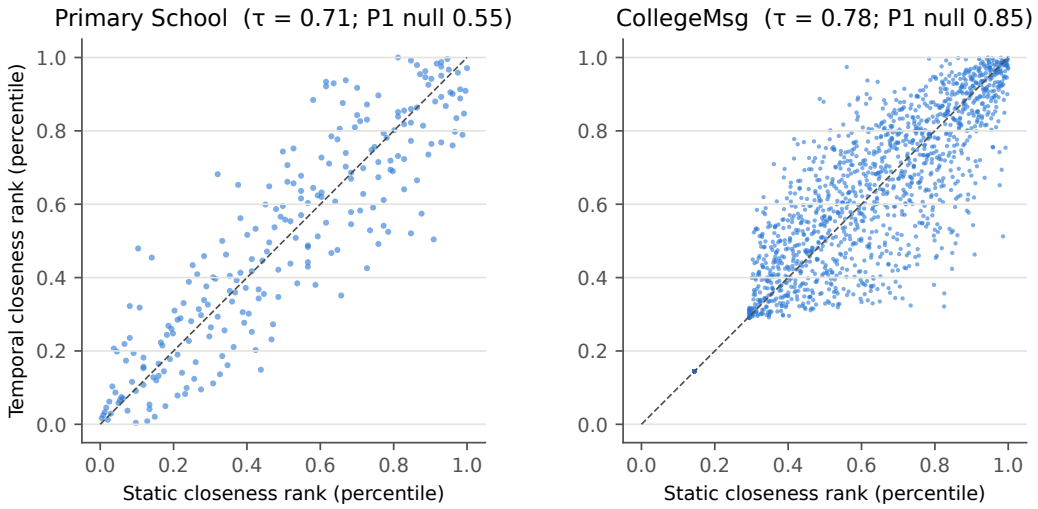


Figure 12. Static (unweighted harmonic) vs. temporal closeness rank of every node; dashed: identical rankings. The tied block at the lower left of the CM panel is the 29% of users who never send a message.

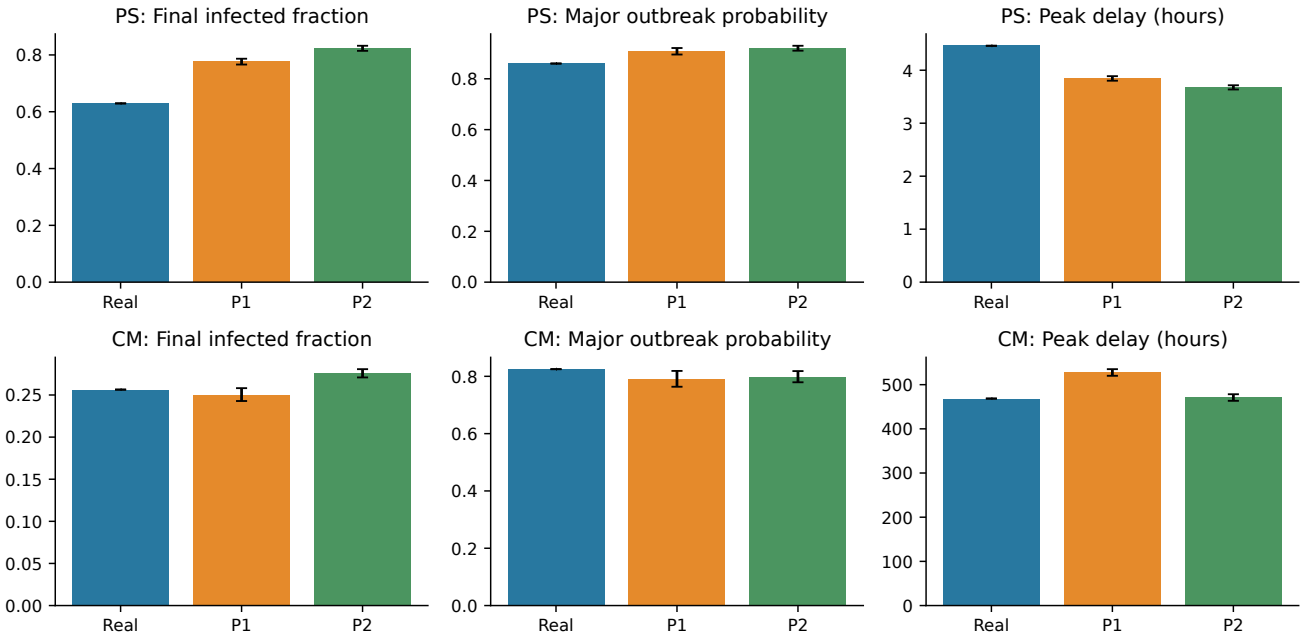


Figure 13. Primary SIR settings. Peak delay is conditional on major outbreaks; errors describe null-network variation.

5.7. Measured cross-network summary

Each row uses the hypothesis-specific measure shown below. Their units and interpretations differ, so the table does not define a single overall sensitivity score.

ID	Measure	PS	CM	CIT
H1	Density range/mean	3.359	3.862	1.798
H2	$B - B_{P1}$	0.016	0.336	–
H3	Adjacent Jaccard	0.220	0.096	–
H4	PA α	–	–	0.714
H5	Adjacent NMI	0.816	0.519	–
H6	Reach ratio R	0.866	0.205	–
H7	Peak ratio vs P1	1.160	0.888	–

Table 12. Measured H1–H7 primary effects; – means inapplicable. H4 uses PA as its scalar representative; ageing, densification and predictive diagnostics retain separate verdicts. H5 uses 15-min PS/7-day CM windows; H7 uses the first parameter setting and major outbreaks.

H1 is supported as window dependence. H2 separates excess burstiness from weaker-than-null memory. H3 and H5 depend on network type and window. H6 shows loss of reachable pairs, with partial ranking disagreement. H7 slowing is supported in primary PS, but not CM. The window grids, node universes, edge semantics and conditioning must accompany these values.

6. Discussion

What the static network already tells us. Conclusions that survive aggregation can be read from Tab. 12. For CIT, the final aggregate describes degree heterogeneity, clustering and connected components. It cannot by itself identify preferential attachment or ageing: these compare new citations with prior degrees and paper ages. Even densification requires a sequence of historical $N(t), E(t)$ pairs, rather than one final graph. For PS and CM, static aggregation preserves the existence of links but hides their timing, repetition and turnover.

What requires time. Some conclusions need windows (Parts I and III) and others the full event order (Parts II and IV). For H2, both PS and CM exhibit higher pooled burstiness than their P1 timestamp-shuffled baselines. The excess is relatively small for PS ($+0.0163$) but much larger for CM ($+0.3361$). However, observed memory is lower than P1 in both networks, so stronger-than-null positive memory is not supported. For H3, adjacent-window link overlap also differs from the randomized baseline, demonstrating that static aggregation cannot capture temporal link turnover. Community reorganisation, time-respecting reachability and spreading speed additionally require temporal information, although their conclusions depend on the corresponding analyses.

Why network types differ. The three networks differ in generating mechanism (physical proximity, voluntary messaging, citing earlier work), time scale and edge semantics (a contact ends, a message is a point event, a citation persists). PS interactions are constrained by school schedules and physical proximity, whereas CM records voluntary online communication over a much longer period. These differences may contribute to the stronger excess burstiness in CM. For H3, PS has lower adjacent-hour Jaccard similarity than P1 (0.2198 versus 0.3668), while CM has higher adjacent-week similarity (0.0958 versus 0.0691). However, the CM contrast reverses at a 14-day window. Thus, temporal persistence depends on both network type and aggregation scale, rather than following a universal pattern. CIT has persistent edges and measured age-dependent target preferences. Direct comparisons remain conditional on differences in event definitions and observation periods.

Robustness. We check the choice of window size, the observation period (first vs. second half), the community-detection resolution and the choice of null model. For H2, PS was evaluated using both raw 20-second contact records and merged contact episodes. Both definitions produce higher burstiness and lower memory than P1, although the excess burstiness changes from $+0.1020$ to $+0.0163$. For H3, PS shows lower link overlap than P1 across all four tested windows, while CM changes from higher to lower overlap as the window increases to 14 days. The P2b endpoint-swap implementation passes structural invariant checks, but its mixing behaviour has not been fully assessed.

For CIT, the full-period densification exponent is 1.343 but falls to 1.065 in the later half, while remaining 1.427 with cutoff 2015. Positive degree preference persists under all declared variants, although its exponent depends on the degree offset. The diameter verdict is unchanged by the

sampled-source sensitivity checks. These are qualitative robustness findings, not evidence that coverage and censoring have been removed.

Citation-specific evidence. CIT supports degree preference and age-dependent target choice, but separates two claims often grouped together: cumulative edges grow superlinearly, while the sampled effective diameter does not shrink. A three-parameter citation curve fits selected histories well, yet its paired MAE advantage over linear prediction is unresolved. Annual zero clustering is imposed by the strict-year snapshot construction. These examples show why representation, fitted-sample description and held-out prediction require separate interpretations. Late reference coverage, within-venue selection and right censoring constrain all citation conclusions.

Limitations. There is one dataset per network type, so generality is limited and conclusions hold "in these data". For PS and CM, pooled burstiness and memory may be influenced by heterogeneous edge activity. P1 preserves the global timestamp distribution and therefore does not eliminate all temporal nonstationarity. The short PS observation period also introduces boundary effects in edge-lifetime measurements. Window-based Jaccard similarity depends on the chosen aggregation scale. Although P2b preserves event participation counts, its endpoint swaps are not guaranteed to sample uniformly from all admissible networks. The 20-draw null ensembles give coarse tail resolution.

CIT also has within-venue sampling, annual timestamp ties and changing reference coverage. Its year-block bootstrap intervals are descriptive; cumulative graph observations are dependent. Forecast selection excludes low-cited papers, and 38.6% of short-history WSB fits hit parameter bounds despite optimizer success in every case; fitted collapse is not independent validation. The primary age-15 target is fully observed, whereas 2020 endpoint comparisons mix forecast horizons. Neither removes within-venue citation-coverage limitations. Boundary effects and sampling bias remain (finite two-day contact window, sensor participation). Citation networks are degenerate temporal networks (DAG, no deletion), which is why several methods do not apply. Each system also contains several time scales [11], so a window that suits one process may hide another. Finally, SocioPatterns data are released under CC BY-NC-SA.

7. Conclusion

The completed citation-network experiments show that the importance of time depends on the question. A final aggregate reveals degree heterogeneity and connectivity, while preferential attachment, ageing and the growth law require historical degrees, publication ages or cumulative snapshots. In the selected AMiner database corpus, sublinear degree preference and age-dependent citation choice coexist with superlinear edge growth, yet the sampled effective diameter does not shrink. WSB compactly describes selected citation histories, but its mean-absolute-error advantage over a simple linear forecast is not established. Thus descriptive fit, temporal mechanism and predictive value should be assessed separately. PS and CM show window-dependent structure, excess burstiness with weaker-than-null pooled memory, and restricted temporal reachability. Static communities conceal PS mixing more clearly than CM reorganisation. Primary PS outbreaks peak later than shuffled sequences, while CM peaks earlier than P1; a universal spreading slowdown is not supported. The hypothesis-specific indicators cannot be combined into a single network-type ranking without additional assumptions. Future work includes broader corpus coverage, more datasets per type, finer-resolution citation dates and predictive tests under alternative censoring and missing-reference assumptions.

References

[1] Albert-László Barabási. The origin of bursts and heavy tails in human dynamics. *Nature*, 435:207–211, 2005.

[2] Albert-László Barabási and Réka Albert. Emergence of scaling in random networks. *Science*, 286(5439):509–512, 1999.

[3] Vincent D. Blondel, Jean-Loup Guillaume, Renaud Lambiotte, and Etienne Lefebvre. Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, page P10008, 2008.

[4] Leon Danon, Albert Díaz-Guilera, Jordi Duch, and Alex Arenas. Comparing community structure identification. *Journal of Statistical Mechanics: Theory and Experiment*, page P09008, 2005.

[5] S. N. Dorogovtsev and J. F. F. Mendes. Evolution of networks with aging of sites. *Physical Review E*, 62:1842–1845, 2000.

[6] Laetitia Gauvin, Mathieu Génois, Márton Karsai, Mikko Kivelä, Taro Takaguchi, Eugenio Valdano, and Christian L. Vestergaard. Randomized reference models for temporal networks. *SIAM Review*, 64(4):763–830, 2022.

[7] Valerio Gemmetto, Alain Barrat, and Ciro Cattuto. Mitigation of infectious disease at school: targeted class closure vs school closure. *BMC Infectious Diseases*, 14:695, 2014.

[8] K.-I. Goh and A.-L. Barabási. Burstiness and memory in complex systems. *EPL (Europhysics Letters)*, 81:48002, 2008.

[9] Derek Greene, Donal Doyle, and Padraig Cunningham. Tracking the evolution of communities in dynamic social networks. In *International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*, pages 176–183, 2010.

[10] Petter Holme. Network reachability of real-world contact sequences. *Physical Review E*, 71:046119, 2005.

[11] Petter Holme. Modern temporal network theory: a colloquium. *The European Physical Journal B*, 88:234, 2015.

[12] Petter Holme and Jari Saramäki. Temporal networks. *Physics Reports*, 519(3):97–125, 2012.

[13] H. Jeong, Z. Néda, and A.-L. Barabási. Measuring preferential attachment in evolving networks. *EPL (Europhysics Letters)*, 61(4):567–572, 2003.

[14] M. Karsai, M. Kivelä, R. K. Pan, K. Kaski, J. Kertész, A.-L. Barabási, and J. Saramäki. Small but slow world: How network topology and burstiness slow down spreading. *Physical Review E*, 83:025102, 2011.

[15] Hyoungshick Kim and Ross Anderson. Temporal node centrality in complex networks. *Physical Review E*, 85:026107, 2012.

[16] Lauri Kovanen, Márton Karsai, Kimmo Kaski, János Kertész, and Jari Saramäki. Temporal motifs in time-dependent networks. *Journal of Statistical Mechanics: Theory and Experiment*, page P11005, 2011.

[17] Gautier Krings, Márton Karsai, Sebastian Bernhardsson, Vincent D. Blondel, and Jari Saramäki. Effects of time window size and placement on the structure of an aggregated communication network. *EPJ Data Science*, 1:4, 2012.

[18] Jure Leskovec, Jon Kleinberg, and Christos Faloutsos. Graphs over time: densification laws, shrinking diameters and possible explanations. In *Proceedings of the 11th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 177–187, 2005.

[19] Jure Leskovec, Jon Kleinberg, and Christos Faloutsos. Graph evolution: densification and shrinking diameters. *ACM Transactions on Knowledge Discovery from Data*, 1(1):2, 2007.

[20] Massimo Marchiori and Vito Latora. Harmony in the small-world. *Physica A*, 285:539–546, 2000.

[21] Naoki Masuda and Renaud Lambiotte. *A Guide to Temporal Networks*. World Scientific, 2016.

[22] Cristopher Moore, Gourab Ghoshal, and M. E. J. Newman. Exact solutions for models of evolving networks with addition and deletion of nodes. *Physical Review E*, 74:036121, 2006.

[23] Peter J. Mucha, Thomas Richardson, Kevin Macon, Mason A. Porter, and Jukka-Pekka Onnela. Community structure in time-dependent, multiscale, and multiplex networks. *Science*, 328(5980):876–878, 2010.

[24] M. E. J. Newman. Clustering and preferential attachment in growing networks. *Physical Review E*, 64:025102, 2001.

[25] Raj Kumar Pan and Jari Saramäki. Path lengths, correlations, and centrality in temporal networks. *Physical Review E*, 84: 016105, 2011.

[26] Pietro Panzarasa, Tore Opsahl, and Kathleen M. Carley. Patterns and dynamics of users' behavior and interaction: Network analysis of an online community. *Journal of the American Society for Information Science and Technology*, 60(5): 911–932, 2009.

[27] Ashwin Paranjape, Austin R. Benson, and Jure Leskovec. Motifs in temporal networks. In *Proceedings of the Tenth ACM International Conference on Web Search and Data Mining (WSDM)*, pages 601–610, 2017.

[28] Nicola Perra, Bruno Gonçalves, Romualdo Pastor-Satorras, and Alessandro Vespignani. Activity driven modeling of time varying networks. *Scientific Reports*, 2:469, 2012.

[29] Jörg Reichardt and Stefan Bornholdt. Statistical mechanics of community detection. *Physical Review E*, 74:016110, 2006.

[30] Bruno Ribeiro, Nicola Perra, and Andrea Baronchelli. Quantifying the effect of temporal resolution on time-varying networks. *Scientific Reports*, 3:3006, 2013.

[31] Giulio Rossetti and Rémy Cazabet. Community discovery in dynamic networks: a survey. *ACM Computing Surveys*, 51 (2):35, 2018.

[32] Martin Rosvall and Carl T. Bergstrom. Mapping change in large networks. *PLoS ONE*, 5(1):e8694, 2010.

[33] Jari Saramäki, E. A. Leicht, Eduardo López, Sam G. B. Roberts, Felix Reed-Tsochas, and Robin I. M. Dunbar. Persistence of social signatures in human communication. *Proceedings of the National Academy of Sciences*, 111(3):942–947, 2014.

[34] Juliette Stehlé, Nicolas Voirin, Alain Barrat, Ciro Cattuto, Lorenzo Isella, Jean-François Pinton, Marco Quaggiotto, Wouter Van den Broeck, Corinne Régis, Bruno Lina, and Philippe Vanhems. High-resolution measurements of face-to-face contact patterns in a primary school. *PLoS ONE*, 6 (8):e23176, 2011.

[35] Jie Tang, Jing Zhang, Limin Yao, Juanzi Li, Li Zhang, and Zhong Su. Arnetminer: extraction and mining of academic social networks. In *Proceedings of the 14th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 990–998, 2008.

[36] V. A. Traag, L. Waltman, and N. J. van Eck. From louvain to leiden: guaranteeing well-connected communities. *Scientific Reports*, 9:5233, 2019.

[37] Dashun Wang, Chaoming Song, and Albert-László Barabási. Quantifying long-term scientific impact. *Science*, 342(6154): 127–132, 2013.